Math 661 Applied Statistics

Module 2 Assignment

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Task 1: Gosset & Guinness — IPS Exercises 1.37, 1.38, 1.39

Context: William Sealy Gosset worked at the Guinness Brewery in Dublin and made substantial contributions to the practice of statistics. In one study, Gosset examined the change in the double stout market before and after World War I (1914–1918). For various regions in England and Scotland, he calculated the ratio of sales in 1925 (after the war) as a percent of sales in 1913 (before the war).

Data: Bristol 94, Cardiff 112, English Agents 78, English O 68, English P 46, English R 111, Glasgow 66, Liverpool 140, London 428, Manchester 190, Newcastle-on-Tyne 118, Scottish 24

Exercise 1.37: Measures of Center

(a) **Mean:**

```
x■ = (1/n) × Σx■ = (1/12) × 1475 = 122.92
```

Calculation: (94 + 112 + 78 + 68 + 46 + 111 + 66 + 140 + 428 + 190 + 118 + 24) / 12 = 122.92

(b) **Median:**

```
Sorted data: 24, 46, 66, 68, 78, 94, 111, 112, 118, 140, 190, 428

With n = 12 (even), median = average of 6th and 7th values

Median = (94 + 111) / 2 = 102.5
```

(c) **Preferred Measure of Center:**

The median (102.5) is preferred over the mean (122.92) because the distribution is strongly right-skewed due to the extreme London value of 428. The mean is pulled upward by this outlier and does not represent the typical sales ratio. The median is resistant to outliers and better represents the center of this distribution. Most regions had sales ratios around 100%, indicating stable or slightly declining sales after WWI.

Exercise 1.38: Measures of Spread

(a) **Standard Deviation:**

```
Formula: s = √[(1/(n-1)) × Σ(x■ - x■)²]

Step 1: Calculate deviations (x■ - x■) where x■ = 122.92

Step 2: Square each deviation

Step 3: Sum of squared deviations = 122962.92

Step 4: s = √(122962.92/11) = 105.73
```

(b) **Quartiles (Using Textbook Method - Median of Halves):**

```
Lower half (first 6 values): 24, 46, 66, 68, 78, 94

Q■ = median of lower half = (66 + 68)/2 = 67.0

Upper half (last 6 values): 111, 112, 118, 140, 190, 428
```

```
Q■ = median of upper half = (118 + 140)/2 = 129.0

Five-Number Summary: Min=24, Q■=67.0, Median=102.5, Q■=129.0, Max=428
```

(c) **Preferred Measure of Spread:**
The IQR (62.0) is preferred over the standard deviation (105.73) because the data contains an extreme outlier (London = 428). The standard deviation, like the mean, is sensitive to outliers. The IQR, measuring the spread of the middle 50% of the data, is resistant to outliers and gives a more representative measure of the typical variability in the dataset.

Exercise 1.39: Outlier Detection and Boxplots

```
(a) IQR: IQR = Q■ - Q■ = 129.0 - 67.0 = 62.0
```

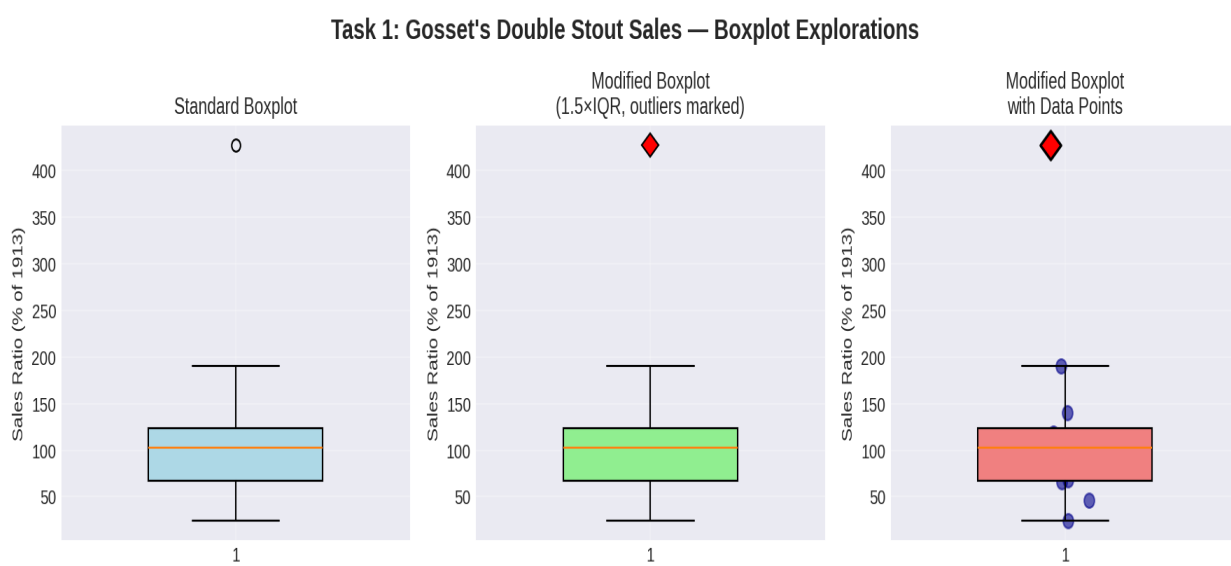
(b) **Outlier Detection (1.5 × IQR Rule):**

```
Lower fence = Q■ - 1.5 × IQR = 67.0 - 1.5 × 62.0 = -26.0

Upper fence = Q■ + 1.5 × IQR = 129.0 + 1.5 × 62.0 = 222.0

Outlier(s): 428 (London = 428 exceeds upper fence of 222.0)
```

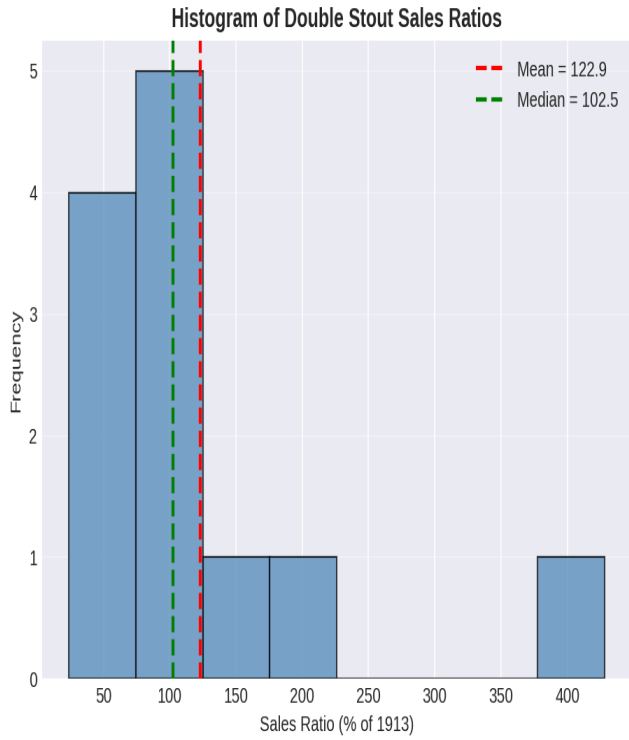
(c) **Boxplot Explorations:**
Below are three boxplot variations exploring different parameter settings. The **standard boxplot** shows the basic five-number summary. The **modified boxplot** explicitly marks outliers beyond 1.5×IQR using a different marker. The **third version** overlays individual data points to show the actual data distribution.



Stemplot

```
Stemplot of Double Stout Sales

Stem | Leaves        Key: 42|8 = 428
-----|---------------
  2  | 4            Stem unit = 10
  4  | 6            Leaf unit = 1
  6  | 6 8
  7  | 8
  9  | 4
 11  | 1 2 8
 14  | 0
 19  | 0
 42  | 8
```



Comparison of Graphical Summaries:

The standard boxplot is compact and shows the five-number summary clearly but does not distinguish outliers visually. The modified boxplot explicitly flags outliers (London at 428) with a different marker, making it more informative for identifying unusual observations. The version with overlaid data points provides the most information, showing both the summary statistics and the actual data distribution. The stemplot reveals the individual data values and confirms the strong right-skew. The histogram with marked mean and median clearly shows the impact of the outlier: the mean (122.9, red dashed line) is pulled far to the right of the median (102.5, green dashed line). For this dataset, the modified boxplot with data points provides the best balance of summary information and transparency.

What I Learned from This Data:

The double stout sales data reveals several important insights about post-WWI market dynamics and the importance of robust statistics: 1. **Outlier Impact:** London's sales ratio of 428% was dramatically different from all other regions. While most regions saw sales remain stable or decline (median ratio ≈ 102%), London experienced over 4× pre-war sales. This single observation strongly influences the mean (122.92) but not the median (102.5), demonstrating why resistant measures are crucial when outliers are present. 2. **Statistical Storytelling:** The wide gap between mean and median (difference of 20.42) is a clear signal of a skewed distribution with outliers. This teaches us to always examine both measures together rather than relying on one alone. 3. **Historical Context:** The data reveals that most UK regions experienced similar post-war patterns in stout consumption, but London was exceptional. This could reflect London's unique position as the capital, potentially benefiting from different economic recovery patterns, population changes, or cultural shifts after WWI. 4. **Practical Data Analysis:** This exercise demonstrates the core workflow of exploratory data analysis:

calculate summary statistics, create multiple visualizations, identify outliers, choose appropriate measures, and interpret findings in context. For decision-making at Guinness, understanding that London was an outlier (not representative of typical performance) would be crucial for strategic planning.

Task 2: Potassium from Potatoes

Context: The 2015 Dietary Guidelines for Americans notes that the average potassium (K) intake for U.S. adults is about half of the recommended amount. A major source of potassium is potatoes. Nutrients in the diet can have different absorption depending on the source. One study looked at absorption of potassium, measured in milligrams (mg), from different sources. Participants ate a controlled diet for five days, and the amount of potassium absorbed was measured. Data for a diet that included 40 milliequivalents (mEq) of potassium were collected from 27 adult subjects.

Data Sample (first 5 subjects): 3096.79 mg, 3607.14 mg, 3584.78 mg, 3256.73 mg, 3286.95 mg, ...
Total sample size: n = 27 subjects

(a) **Stemplot of Potassium Absorption Data:**
The stemplot groups data by thousands (stem) and hundreds (leaf):

```
Stemplot of Potassium Absorption (mg)

Stem | Leaves         Key: 26|6 = 2660 mg
-----|---------------
     |               Stem unit = 1000 mg
     |               Leaf unit = 100 mg
-----|---------------
  2  | 6 7 9 9 9 9
  3  | 0 0 0 0 0 1 1 1 1 2 2 2 2 2 2 4 5 5 5 6
  4  | 2
```

(b) **Distribution Pattern:**
The distribution of potassium absorption shows a roughly **unimodal** pattern with the bulk of observations concentrated between approximately 2,950 mg and 3,300 mg. The distribution appears **slightly right-skewed**, with a longer tail extending toward higher values (3,500–4,200 mg). Most of the data clusters around 3,000–3,200 mg, suggesting that this is the typical range of potassium absorption from the potato-based diet. The peak concentration in the 3,000–3,300 mg range indicates relatively consistent absorption among most participants, with some individual variation at the extremes.

(c) **Outlier Analysis (Showing All Working):**

Step 1: Calculate Quartiles

```
Sorted data: n = 27 observations

Lower half (first 14 values): median = Q■ = 3036.62 mg

Upper half (last 14 values): median = Q■ = 3283.06 mg
```

Step 2: Calculate IQR

```
IQR = Q■ - Q■ = 3283.06 - 3036.62 = 246.44 mg
```

Step 3: Calculate Fences (1.5 × IQR Rule)

```
Lower fence = Q■ - 1.5 × IQR = 3036.62 - 1.5 × 246.44 = 2666.95 mg

Upper fence = Q■ + 1.5 × IQR = 3283.06 + 1.5 × 246.44 = 3652.73 mg
```

Step 4: Identify Outliers

```
Values below 2666.95: 2664.38

Values above 3652.73: 4213.49
```

Interpretation: Two outliers were identified: 2664.38 mg (below lower fence) and 4213.49 mg (above upper fence). The low value of 2664.38 mg is only marginally below the lower fence (by 2.57 mg), suggesting relatively low but not extremely unusual absorption for this participant. In contrast, the high value of 4213.49 mg is substantially above the upper fence (by 560.76 mg), representing unusually high potassium absorption. This could reflect individual physiological differences in absorption efficiency, measurement variation, or other participant-specific factors.

(d) **Shape, Center, and Spread (Complete Summary):**

Shape:

The distribution is **slightly right-skewed** and approximately **unimodal**. Evidence of right-skew: (1) Mean (3208.44 mg) > Median (3130.37 mg), difference = 78.07 mg; (2) Presence of high outlier pulling the tail rightward; (3) Visual inspection of stemplot and histogram shows longer right tail.

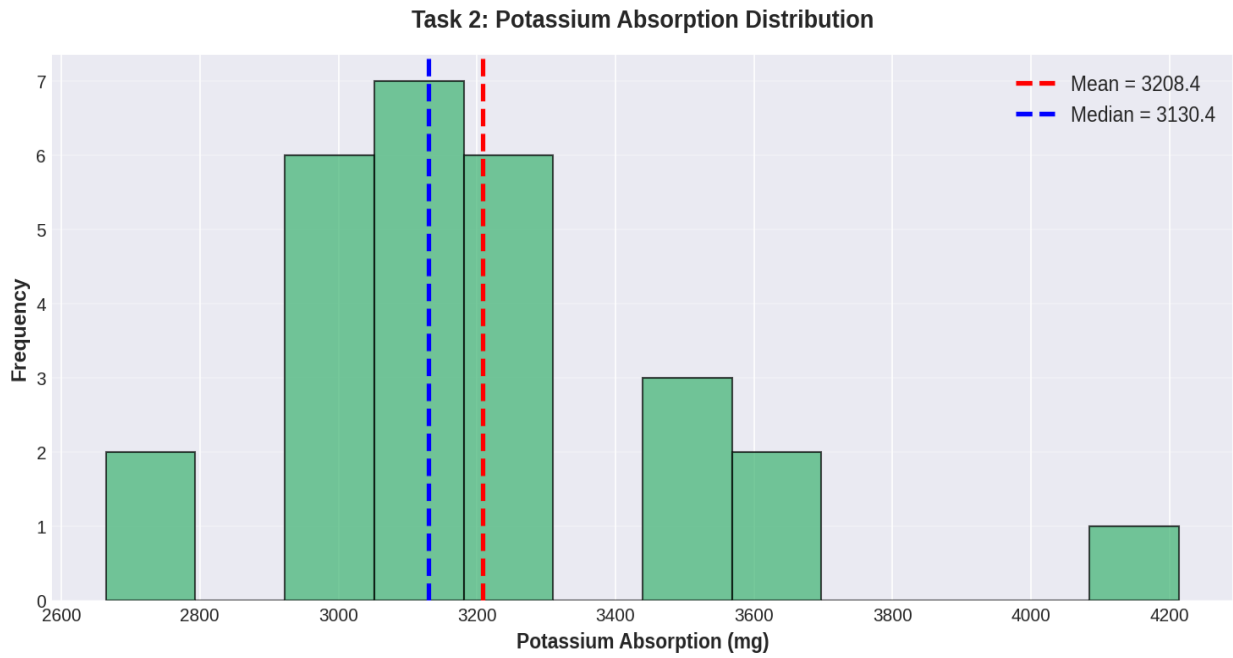
Center:

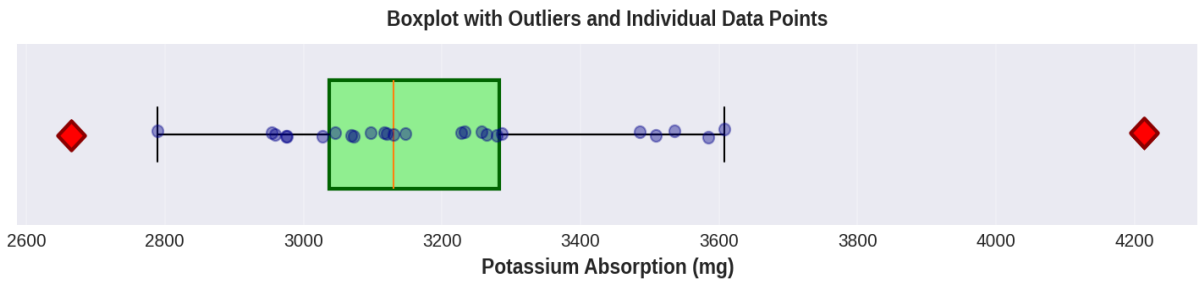
Mean = 3208.44 mg (calculated as Σx■/n = 86627.81/27)
Median = 3130.37 mg (middle value of sorted data)
The relatively small difference (78.07 mg, or 2.5% of median) suggests the skew is moderate rather than extreme.

Spread:

Standard Deviation (s) = 306.68 mg
Calculation: s = √[Σ(x■ - x■)²/(n-1)] = √[2445311.84/26] = 306.68 mg
IQR = 246.44 mg (middle 50% spread)
Range = 1549.11 mg (from 2664.38 to 4213.49)
Coefficient of Variation = 9.6% (measures relative spread)

Five-Number Summary: Min=2664.38, Q■=3036.62, Median=3130.37, Q■=3283.06, Max=4213.49





Conclusion: The potassium absorption data reveals relatively consistent absorption among participants (median ≈ 3,130 mg), with most values falling within a fairly narrow range. The slight right-skew indicates some participants (like the 4,213 mg outlier) absorb substantially more potassium than typical, while absorption below 2,700 mg is rare. For dietary recommendations, this variability (CV ≈ 9.6%) suggests that individual responses to potato-based potassium sources can differ meaningfully, though most people fall within a predictable range.

Task 3: Trimmed Mean — IPS Exercise 1.59

(Practice Task - Not Graded)

A **trimmed mean** is computed by removing a fixed percentage of observations from each end of the sorted data before computing the mean. This provides a measure of center that is resistant to outliers, offering a compromise between the mean (sensitive to outliers) and the median (completely resistant but uses less information).

Double Stout Data (Task 1):

```
Original Mean: 122.92 | Median: 102.5
5% Trimmed Mean: 102.30
10% Trimmed Mean: 102.30
```

For n=12, 5% trim removes 0.6 observations from each end (rounds to 1 each), and 10% trim removes 1.2 observations (also rounds to 1 each in conservative calculation). The 5% and 10% trimmed means (102.30 and 102.30) are substantially lower than the full mean (122.92) and very close to the median (102.5). This confirms that removing the London outlier brings the mean much closer to the typical value, demonstrating the outlier's strong influence on the untrimmed mean.

Potassium Data (Task 2):

```
Original Mean: 3208.44 | Median: 3130.37
5% Trimmed Mean: 3190.00
10% Trimmed Mean: 3181.65
```

For the potassium data (n=27), the trimmed means (3190.00 and 3181.65) are very close to both the full mean (3208.44) and the median (3130.37). This indicates that this distribution has fewer influential extreme values compared to the double stout data. The potassium data is more symmetric with less extreme outlier impact.

Task 4: Proof — Deviations from the Mean Sum to Zero

(Practice Task - Not Graded)

Theorem: For any set of observations $x_1, x_2, ..., x_n$ with mean $\bar{x}$, the sum of deviations from the mean equals zero: $\Sigma(x_i - \bar{x}) = 0$

Proof: By definition, the sample mean is: $\bar{x} = (1/n) \times \Sigma x_i$ Multiplying both sides by n: $n \times \bar{x} = \Sigma x_i$ Now consider the sum of deviations: $\Sigma(x_i - \bar{x}) = \Sigma x_i - \Sigma \bar{x} = \Sigma x_i - n \times \bar{x}$ [since $\Sigma \bar{x}$ sums $\bar{x}$ n times] $= \Sigma x_i - \Sigma x_i$ [substituting $n \times \bar{x} = \Sigma x_i$] $= 0$ ■ **Significance:** This mathematical property explains why variance and standard deviation use **squared** deviations rather than raw deviations. Since raw deviations $(x_i - \bar{x})$ always sum to zero, they cannot serve as a useful measure of spread—positive and negative deviations cancel out. Squaring the deviations makes all terms positive, allowing us to measure the total variability in the data.

Numerical Example:

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
2	-4.0	16.0
4	-2.0	4.0
5	-1.0	1.0
7	+1.0	1.0
12	+6.0	36.0
Sum:	0.0	58.0
Mean:	$\bar{x} = 6.0$	$s^2 = 14.5$

This example demonstrates that while $\Sigma(x_i - \bar{x}) = 0.0$, the sum of **squared** deviations is 58.0, which gives us meaningful information about spread.

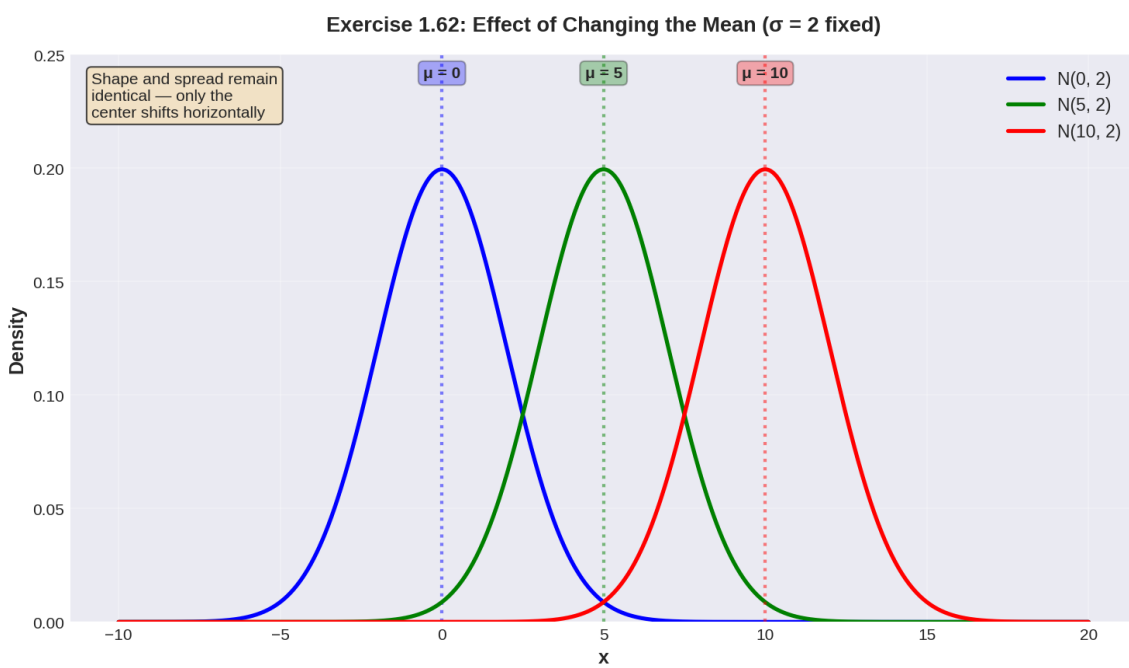
Task 5: Sketching Density Curves — IPS Exercises 1.62, 1.63, 1.64

Task Goal: Create simple conceptual sketches showing how normal density curves change when parameters are modified. These are **hand-drawn style** sketches to demonstrate understanding of concepts, not polished computer graphics.

Normal Distribution Fundamentals: A normal density curve is completely determined by two parameters: the mean μ (which controls the center/location) and the standard deviation σ (which controls the spread/width). The total area under any density curve equals 1.

Exercise 1.62: Effect of Changing the Mean (σ fixed)

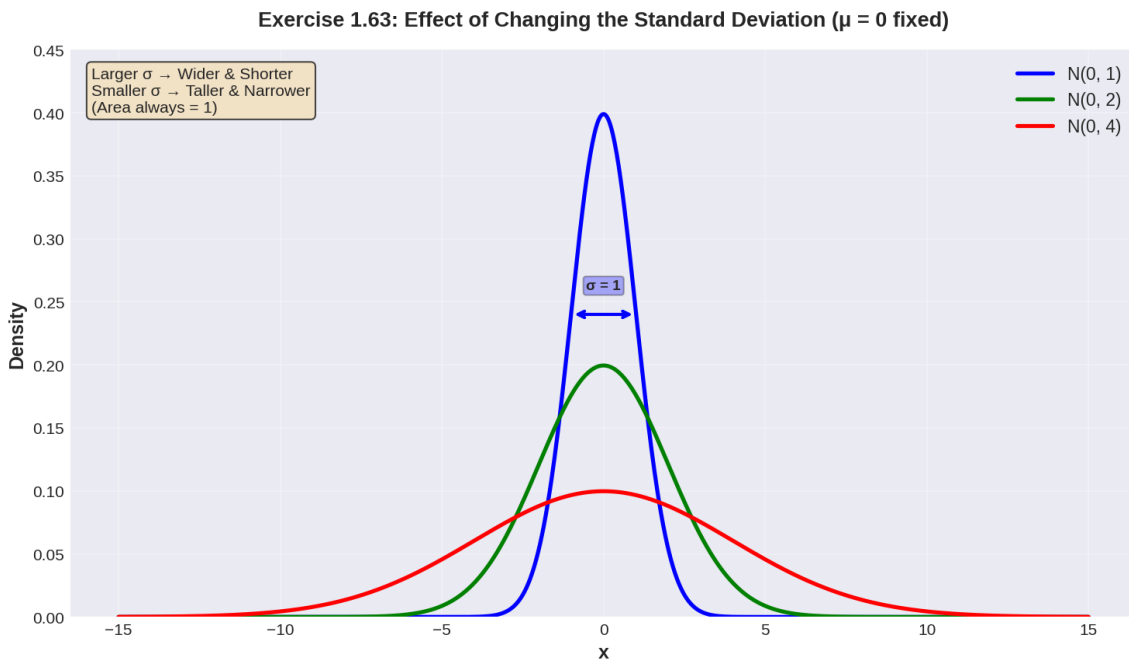
Concept: When we increase the mean μ while holding σ fixed, the entire curve **shifts horizontally** along the x-axis. The shape and spread of the curve remain identical—only the center moves. Below shows three normal curves all with $\sigma = 2$, but with means $\mu = 0$, $\mu = 5$, and $\mu = 10$.



Exercise 1.63: Effect of Changing the Standard Deviation (μ fixed)

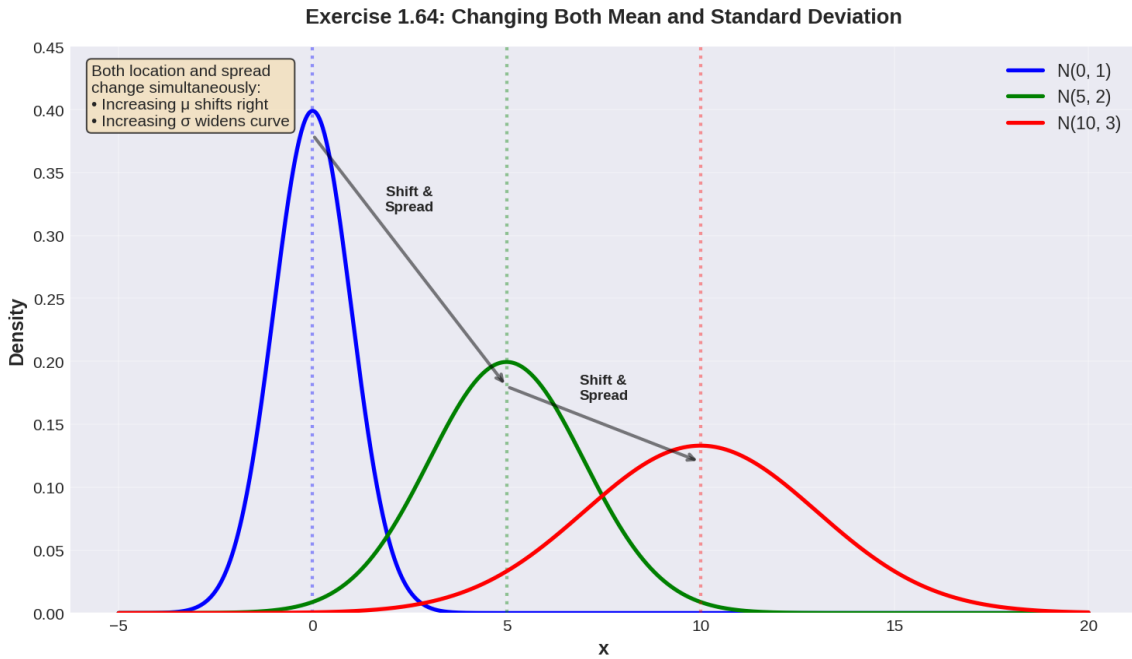
Concept: When we increase the standard deviation σ while holding μ fixed, the curve becomes **wider and shorter** (flatter). Since the total area under any density curve must equal 1, a wider curve must be proportionally shorter. A smaller σ produces a taller, narrower curve concentrated near the mean. Below shows three normal curves all with

$\mu = 0$, but with standard deviations $\sigma = 1$, $\sigma = 2$, and $\sigma = 4$.



Exercise 1.64: Changing Both Mean and Standard Deviation

Concept: When both parameters change simultaneously, the curve both **shifts location** (due to changing μ) and **changes spread** (due to changing σ). The combined effect is that curves differ in both center and shape. Below shows three curves: N(0,1), N(5,2), and N(10,3), demonstrating progressive shifts rightward and increasing spread.



Key Insights: These conceptual sketches demonstrate the fundamental properties of normal distributions. Changes in μ cause horizontal shifts without affecting shape, while changes in σ cause vertical and horizontal scaling (wider/narrower) while maintaining the bell shape. Understanding these transformations is crucial for working with normal models in practice, where different phenomena may have the same shape but different centers or spreads.